

Computing results and running analyses

Table of contents

1	Introduction	2
2	Background	3
2.1	Coreference resolution	3
2.2	Three kinds of <i>they</i>	3
2.3	LLMs as (linguistic) annotators	5
3	Data	5
3.1	Participants	6
3.1.1	Demographics	6
3.1.2	Expertise	6
3.2	LLMs	6
3.2.1	Models	6
3.2.2	Prompting Strategies	7
4	Methods	7
5	Results	7
5.1	Hypothesis 1	7
5.2	Hypothesis 2	8
5.3	Results by label	10
5.3.1	Singular vs Plural	10
5.3.2	Context-(in)dependence	12
5.3.3	Performance with innovative forms of <i>they</i>	14
5.3.4	Projection onto naturalistic data	14
5.3.5	Cost	15
6	Discussion	16
6.1	L1 Crowdworkers vs Local L2 Students	16
6.2	Hypothesis 2	17

6.3 Hypothesis 3	18
6.4 Research question 2	18
6.5 Limitations	19
7 Conclusion	19
7.1 Debugging etc.	20
References	20

- let GPT tell me when it accessed the context
- “treat different examples of same type as experimental condition” – sounds good, figure out if makes sense in my case
- Check for other publications on LLM annotations and see if journals apply; otherwise Corpus Linguistics and Linguistic Theory
- Find out what LLMs have been used for in linguistics and/or NLP for introduction
- Write texts for results section
- Expand notes into text chunks
- Background on *they*
- Background on LLMs as annotators
- Conclusion and Intro
- Choose journal and format accordingly
- Tutorial on exporting to Word
- Quick copy-editing by Jessica

1 Introduction

Large Language Models (LLMs) have not only permeated wider society, they have also begun changing the methodological landscape in the humanities and linguistics more specifically. The promise of the team behind GPT-3 was that Natural Language Processing (NLP) as a whole would shift “to using task-agnostic pre-training and task-agnostic architectures” Brown et al. (2020, 1), i.e. general-purpose language models able to perform a wide range of tasks without task-specific training. Indeed, LLMs have found a wide variety of uses: LLMs have been used for X, Y, and Z in linguistics where their performance was found to be so and so.

The present work was performed in the context of an older effort starting in my Master’s thesis (2022). The then-new GPT-3 was tested for its performance at disambiguation uses of *they* between plural and singular (see below). Coreference resolution as a field precedes LLMs, though it was always limited by the many challenges posed by the nature of coreference, such

as real-world knowledge required for textual understanding and context-dependency. What is more, English pronouns have shifted in recent years, most notably with the extension of *they* as a personal pronoun for people of non-binary gender, but also with the growing visibility of neologistic pronouns. LLMs showed promise because they exhibited context-sensitive performance in several areas. While its performance with singular *they* in the dataset of the Master’s thesis was unsatisfactory, the same test was later repeated using newer models. When Anthropic’s LLM Claude Sonnet reached both recall and precision of over 80% on the same dataset in late 2024, it was hypothesized that LLMs had reached human-like performance in this task. To assess this reliably and accurately, the Research Question was formulated: “To what extent are number-annotations of authentic uses of *they* by LLMs reliable when compared to trained human annotators?”

To answer this question, three levels of performance quality were defined, as expressed by the following three hypotheses:

- Hypothesis 1: The most accurate number-disambiguator of *they* is going to be an LLM.
- Hypothesis 2: The most accurate LLM at number-disambiguating *they* is going to perform at a higher accuracy than the mean human accuracy.
- Hypothesis 3: The performance of the most accurate LLM is not going to be significantly worse at rare and innovative uses of *they* than the mean human performance.

The three hypotheses were defined to capture the extent to which LLM performance could potentially be comparable to that of humans. Even if Hypothesis 1 is not confirmed by the results, hypothesis 2 would still attest that LLMs largely outperform humans. If that were not confirmed either, hypothesis 3 would still justify the use of LLMs for semantically-complex tasks such as resolution of innovative pronouns. Additionally, the present work also allowed to assess more generally and qualitatively in what ways LLMs are human-like and in what ways they are not; and if, perhaps, any revealing patterns could be found in the recorded behaviors of LLMs and humans. This is expressed in the additional research question: “In what ways are LLMs human-like and in what ways are they not when it comes to number-disambiguation of *they*?”

2 Background

Point out that the term LLMs is not synonymous with decoder-only models but that the former will be used for simplicity and because the models tested in this work are all decoder-only. (ref to Arslan, Erol, and Eryigit 2026)

As this task is supposed to represent any task relevant in the field of linguistics in which the mere surface level is not enough for correct processing, the following background is not strictly relevant. It will help to contextualize the present research and justify the task design.

- Haq et al. (2025) also employ commercial and non-commercial LLMs. They don't show that either is inherently better but that model size is an important factor. Commercial models tend to be larger as they are trained on more data, though open-weight models have caught up in that regard (# SOURCE).
- Arslan, Erol, and Eryigit 2026 investigate Coreference Resolution with LLMs and fine-tune models for it. => I do not even fine-tune and reach human-like results They investigate zero anaphors (i.e. omitted pronouns) and find that LLMs perform well on them, which is relevant for the present research as it shows that LLMs show "a deep understanding of the contextual focus". Their work crosses linguistic boundaries which poses interesting challenges. My work does not do that but instead contains innovative language which tends to be rare in more general-purpose datasets.

2.1 Coreference resolution

Describe task

Point out that in this case, the mention detection (ref to textbook) is precluded by our interest in the pronoun *they* and that the task is thus reduced to a classification task of number-disambiguation of *they*.

2.2 Three kinds of *they*

(plural, generic, singular)

The pronoun *they* is generally used as a plural pronoun of the third person, meaning that it refers to groups of people or things that are not directly addressed. However, it is rather well-known that its usage is not limited to plural reference but that a "singular *they*" exists as well. The reason that this grammatical ambiguity is so well-known is that feminist discourses of the latter half of the 20th century drew attention to contexts in which writers would use the pronoun *he* despite the context not strictly mandating a masculine pronoun. This is known as "generic *he*". To avoid linguistic and thus cognitive erasure of women, the use of the *they* in such contexts was advocated instead. This discourse seamlessly flowed into a new era of gender politics that drew attention to at least two new pronoun-related concerns: 1) the use of *they* to mask gender wherever it is not strictly necessary to reveal it, and 2) the use of *they* as a personal pronoun for people of non-binary gender. The first concern is perhaps not as prominent as the second, and speakers are perhaps not as aware of it. The following examples illustrate how this tendency reveals itself:

#TODO EXAMPLES

Example N comes from a public figure in science communication of mathematics, Matt Parker. It is perhaps surprising that he avoids using the gendered pronoun *he* in reference to a known male mathematician. In Example M, Parker avoids using a gendered pronoun when talking

about a person who the audience has not seen yet, and therefore does not know their gender. In my personal correspondence with Parker, he revealed that this was indeed a deliberate choice that he decided to make in his public speaking, specifically “to avoid creating expectations about [...]”. This is a clear example of the first concern mentioned above, i.e. the use of *they* to mask gender. The second concern is more straightforward and is best explained by linguists and other writers who are more directly involved in the discourse around non-binary gender, such as X and Y.

What both concerns mean for the present research is that they make the use of truly singular, and not merely generic, *they* more likely. It can also, and perhaps more usefully, be called “referential singular *they*”. It was alluded to above that *they* was propagated to replace generic *he* in the second half of the 20th century, and indeed it has been found that *they* has since expanded to more contexts that had not been considered before (#REFS). To better grasp the changing use of *they*, synchronically and diachronically, it is thus necessary to highlight the distinction between these two innovations of *they* use.

Generic *they* can be conceived as... #TODO

Singular *they* can be conceived as... #TODO

While this distinction goes beyond the traditional dichotomy of singular and plural, it falls short of the much more fine-grained distinctions proposed in the literature, most prominently in Newman’s work (1994, 1997).

- generic ‘they’ is older; it may thus have reached the use of people who oppose singular *they*
- singular *they* is rarer than generic and much rarer than plural. Further distinctions are relevant and necessary; they can and should be made when investigating usage samples up-close. What is being developed here is a way to efficiently collect instances of generic and singular usage in useful amounts, but leaving the fine-grained analysis to future work.

2.3 LLMs as (linguistic) annotators

3 Data

First, we need to import all raw data by calling the relevant script. The raw data are shaped like so:

- Human participants have a dictionary each in which every data point, their annotation, and (optional) their comment is found.
- Human participants are put into 5 blocks, blocks A-D and the group of pilot participants
- LLMs come in three different groups: commercial, open source, and local

- in each LLM group, there are different combinations of LLMs and prompting strategies, of which several runs may exist
 - each combination of LLM and prompting strategy is treated as a participant.
 - For every run of that combination, a dictionary is found in which every data point is found with their annotation and their full response saved like human annotators' comments
- Point about MA-era-dataset being limited (number of singular they examples; context-independence)

174 authentic uses of they (Reddit) Context-(in)dependent 56–59 for plural/generic/singular (do this dynamically & as a table)

Each annotation instance was treated as an item-level trial for scoring and condition-wise breakdowns, while inferential comparisons were conducted on aggregated participant or run-level performance

Mention 10 intro examples that were cherry-picked for balance and had a precise reasoning behind the solution and were identical for all human participants. Clarify if those 10 examples were part of the test set for LLMs (shouldn't have been but need to ensure).

Randomly sampled for each participant to avoid bias

System of blocks with overlap

Point out that dataset is available

To illustrate, the variable name `commercial_llm_responses` has the entry “context-agnostic_zero-shot_claude-opus-4-1-20250805”, which tells us the prompting strategy (context-agnostic-zero-shot) and the LLM with its dated version (claude-opus-4-1-20250805). This entry has three runs saved. Each run is of the type “dict” and has 174 data points saved with their answers as keys and values. This number corresponds to the number of data points in the original test set.

3.1 Participants

Human annotators are organized in four different blocks due to limits to study duration, i.e. avoiding influence from participant fatigue. Instead of testing everyone with the full 174 data points from the test set, every block of participants received a shared set of data points that was identical among all of them and one part which was only solved by their respective block. The distribution of examples among the blocks was done randomly while ensuring within-block balance of antecedent types.

Blocks A-D have 11, 12, 10, and 8 participants, respectively. The piloting group had 7 participants. In terms of data points, every participant regardless of their block received 48 examples to annotate. This ensured a study duration of roughly one hour per participant.

3.1.1 Demographics

Women, age range (+ bibliographic justification)

3.1.2 Expertise

Explain issue with Prolific setup. Explain possibility of post-hoc culling the participants by attentiveness to still get appropriate data.

3.2 LLMs

3.2.1 Models

(so far) only the largest commercially available LLMs from the Claude & GPT family

Open Source:

Used this as my main guiding resource to tick the most important boxes: <https://www.bentoml.com/blog/navigating-the-world-of-open-source-large-language-models>

So far run: - Deepseek latest few models - Qwen3 latest few models - Momo V2 Flash (only non-free available on OpenRouter)

Fully local models make somewhat less sense for this kind of “piloting for the field” experiments as researchers having access to machines with 20+GB of RAM is less realistic than researchers getting access to computing clusters that can run far larger models. The latter is a more typical setup that you will see at research institutions and is often more readily financed than a machine that is only used by one person and costs double or triple of what their allocated funds typically allow.

Hence, the “local” category will comprise open source models run in a university-run computing cluster.

3.2.2 Prompting Strategies

Context-agnostic or with URL which only works for commercial models.

Invented examples were used as dev set to develop a prompting strategy.

The task description has brief examples interspersed, not fully-fledged ones.

Structured output reduced to just the last word. Worked perfectly.

4 Methods

Explain basic GT comparison

Justify choice of accuracy (F1 doesn't necessarily make sense over accuracy as the three classes are balanced. Also mention that MCC was considered.)

Confirmatory and quasi-experimental study

5 Results

5.1 Hypothesis 1

The first hypothesis aims to test if any LLM-based approach would yield the highest overall performance.

Table 1: The ten highest-performing participants by accuracy

Participant Group	Accuracy
block_Pilot	96%
context-via-permalink_google-gemini-3-pro-preview	92%
context-via-permalink_gpt-5	91%
context-ondemand_zero-shot_gpt-5	90%
block_A	90%
block_B	90%
block_B	90%
block_C	90%
block_D	90%
block_D	90%
context-agnostic_zero-shot_gpt-5	89%
context-via-permalink_gpt-5-2025-08-07_with-justification-unspecified	88%
block_C	88%
block_Pilot	88%
Non-Linguist Human Nr. 18	88%
context-via-permalink_deepseek-deepseek-v3.2-speciale	87%
context-via-permalink_claude-opus-4-1-20250805	86%
context-agnostic_zero-shot_z-ai-glm-5.1	86%
majority_vote	86%
context-via-permalink_gpt-5-2025-08-07_with-justification-conservative	86%
context-agnostic_zero-shot_claude-opus-4-1-20250805	85%
block_B	85%
context-via-permalink_claude-sonnet-4-20250514	84%

Table 1: The ten highest-performing participants by accuracy

Participant Group	Accuracy
context-via-permalink_z-ai-glm-4.7	84%
block_B	83%

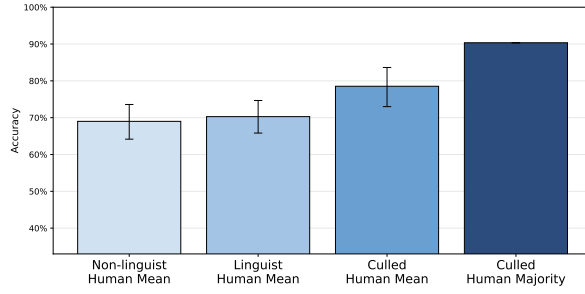
5.2 Hypothesis 2

Table 2: Overall accuracy by three human groups

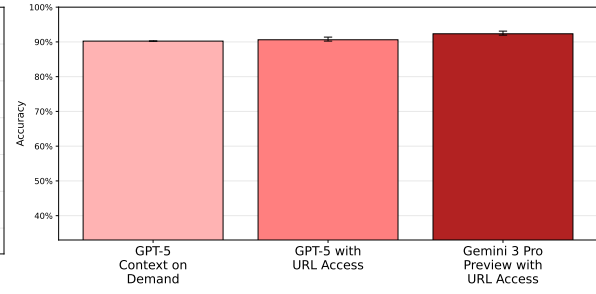
Group	Accuracy
Non-Pilot Humans	68.5%
Pilot Humans	80.8%
Non-Linguists	69.0%

The difference between the crowdworkers and the local student pilot participants is not statistically significant ($p = 0.059$, $\alpha = 0.05$). For inferential testing in this section, only Non-Pilot Humans and Pilot Humans were compared. The Non-Linguists group is reported descriptively for context.

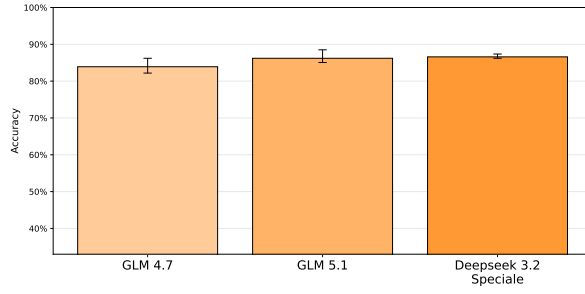
The group of culled participants has 20 participants, of which two are local students and 18 are crowdworkers.



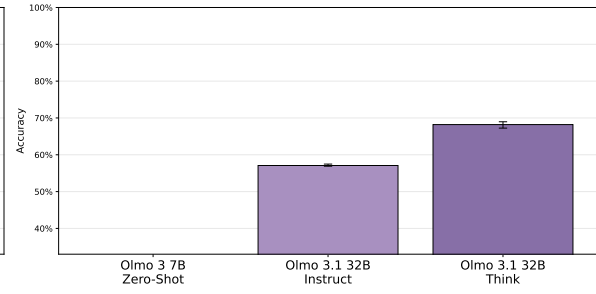
(a) Human Group Accuracies



(b) Commercial LLM Accuracies



(c) Open Source LLM Accuracies



(d) Local LLM Accuracies

Figure 1: Participant accuracies with 95% confidence interval

5.3 Results by label

5.3.1 Singular vs Plural

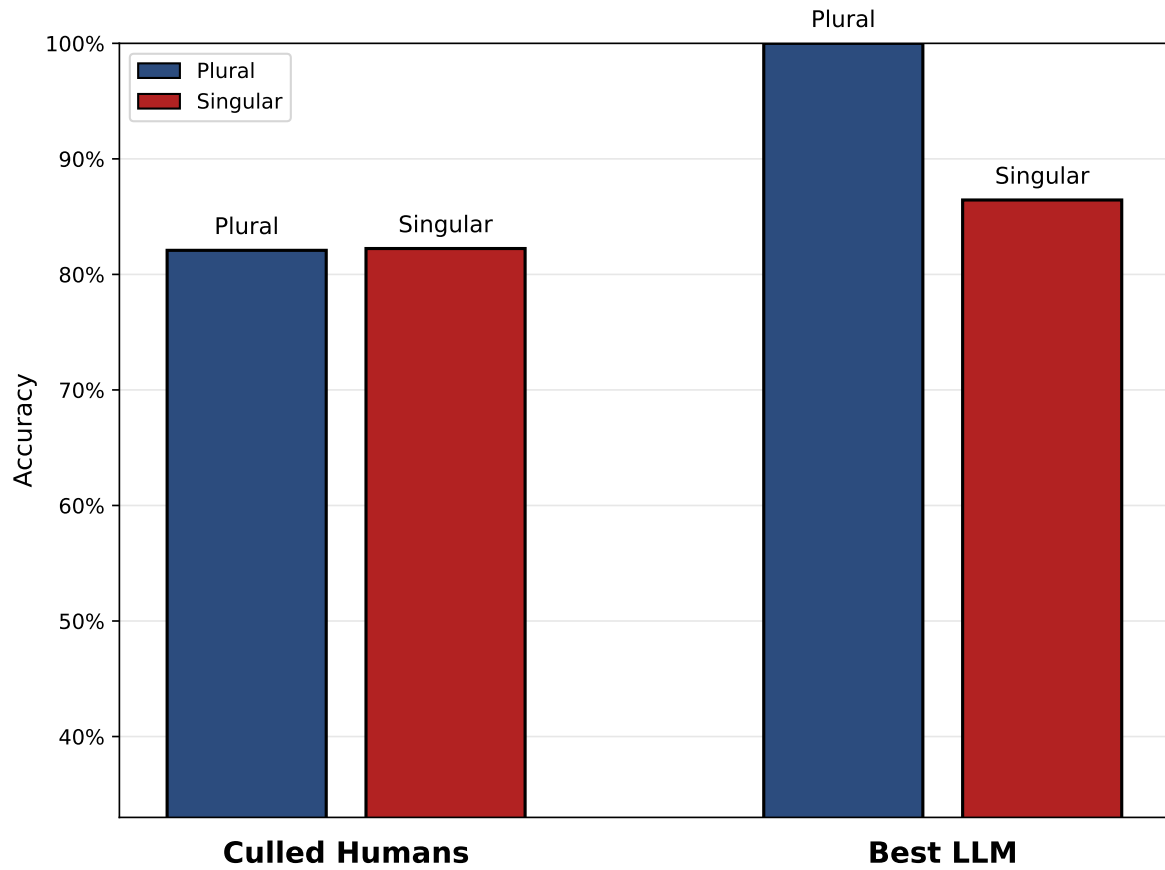


Figure 2: Accuracy difference between plural/singular *they* and LLMs/humans

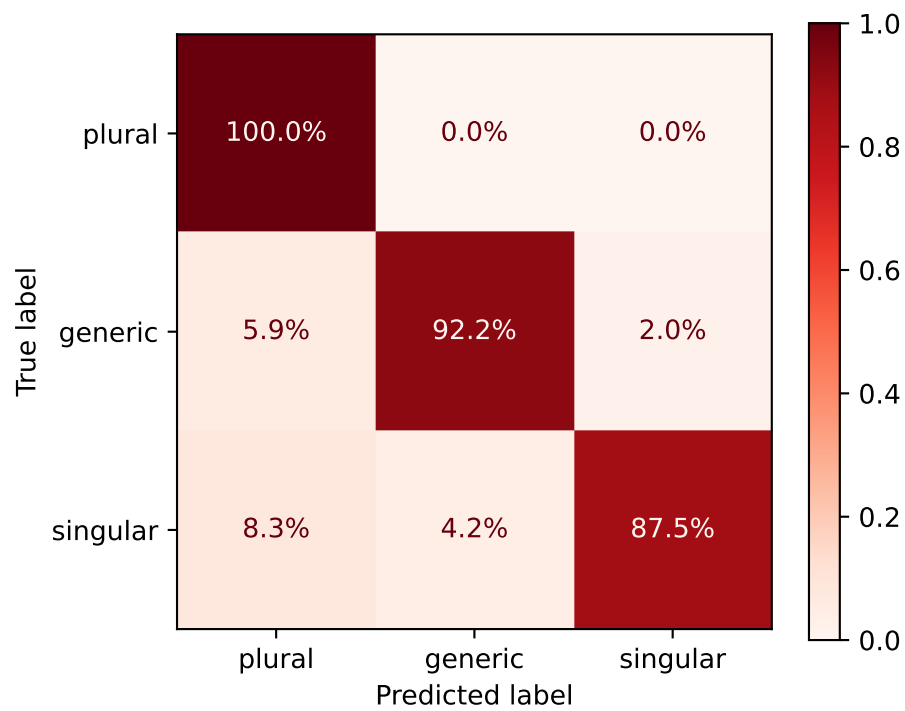


Figure 3: Confusion matrix for the best LLM

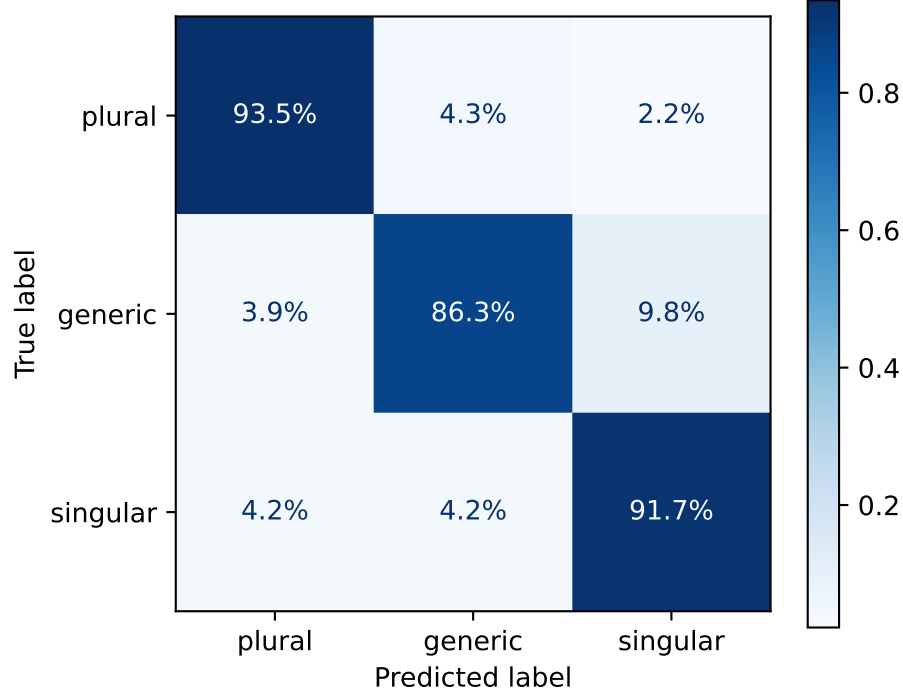


Figure 4: Confusion matrix for culled human mean

TOST (independent samples, Welch)

- $\alpha = 0.050$, $\Delta = 0.020$
- $t_{\text{lower}}=2.232$, $p_{\text{lower}}=0.0189$ (NI decision uses this)
- $t_{\text{upper}}=0.792$, $p_{\text{upper}}=0.7810$

Decision: Non-inferiority **established**.

5.3.2 Context-(in)dependence

Table 3: Accuracy for context-dependent and context-independent items for two different LLM strategies and the culled human mean

Group	Context-Dependent	Context-Independent
Culled Human Mean	90.0%	77.5%
GPT-5 Context-Agnostic	79.4%	92.9%
GPT-5 with URL Access	83.3%	94.5%

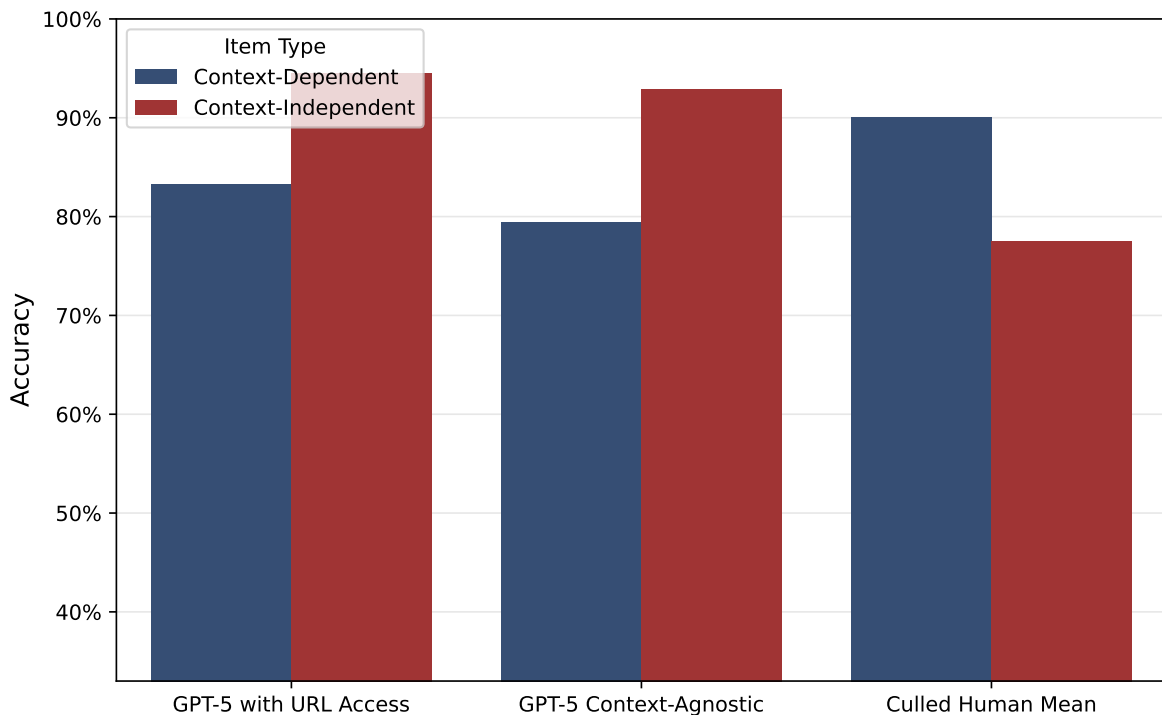


Figure 5: Accuracy for context-dependent and context-independent items for two different LLM strategies and the culled human mean

When looking at the behavior of GPT-5 with URL access, it shows a difference between my own expectations of what examples required context for disambiguation and which ones did not. Of the 34 items that I marked as context-dependent, less than half (14) caused the model to access the context via provided URL. In turn, of the 25 items that did cause the model to check context, only 14 were marked as context-dependent by me. Among the cases that the model failed to look up context I had marked as necessary, however, its accuracy at disambiguating the meaning of *they* still lay at a relatively high 80.0%, which is very close to the average accuracy of my local study pilots over all items, 80.8%. Among the cases that the model had correctly decided to look up context, its accuracy was only 76%, showing that it likely lacks capabilities in navigating heterogenous and complex contexts. This means that the model’s behavior does not fully align with my expectations of what items require context for disambiguation, and it also accesses context for some items that I would have considered context-independent. It is not clear what causes this difference in behavior but it does point to LLMs’ reasoning capabilities being different from human reasoning, generally to the detriment of the results. In light of study design, it suggests that LLMs fare better without having to research information on their own in order to complete a task, as noticing the lack of necessary information seems to be still too great a challenge for such models. This may impede the study of especially complex linguistic phenomena that require contextual and

semantic understanding of text.

When not specifying that the model should access context only in doubtful cases, thus leaving it up to the model, there is a clear tendency for the model to access context more often. Namely, of the 34 context-dependent cases 28 were checked for context, which is a recall of 82%. However, the model checked a total of 106 cases for their context, of which only 28 were actually context-dependent, meaning that the precision of the model's decision to check context was only 26%. This means that while the model is relatively good at recalling cases that require context, it also checks for context in many cases where it is not necessary, which may lead to worse performance due to the model's difficulties in navigating complex contexts. Indeed, in the specific context of this study, there was a measurable drop in accuracy by 10% (from 94% to 84%) when the model accessed context vs when it did not. For this reason, the same model performed better when it did not have the option to access context (89%) than when it did (86%), as shown in the previous section.

5.3.3 Performance with innovative forms of *they*

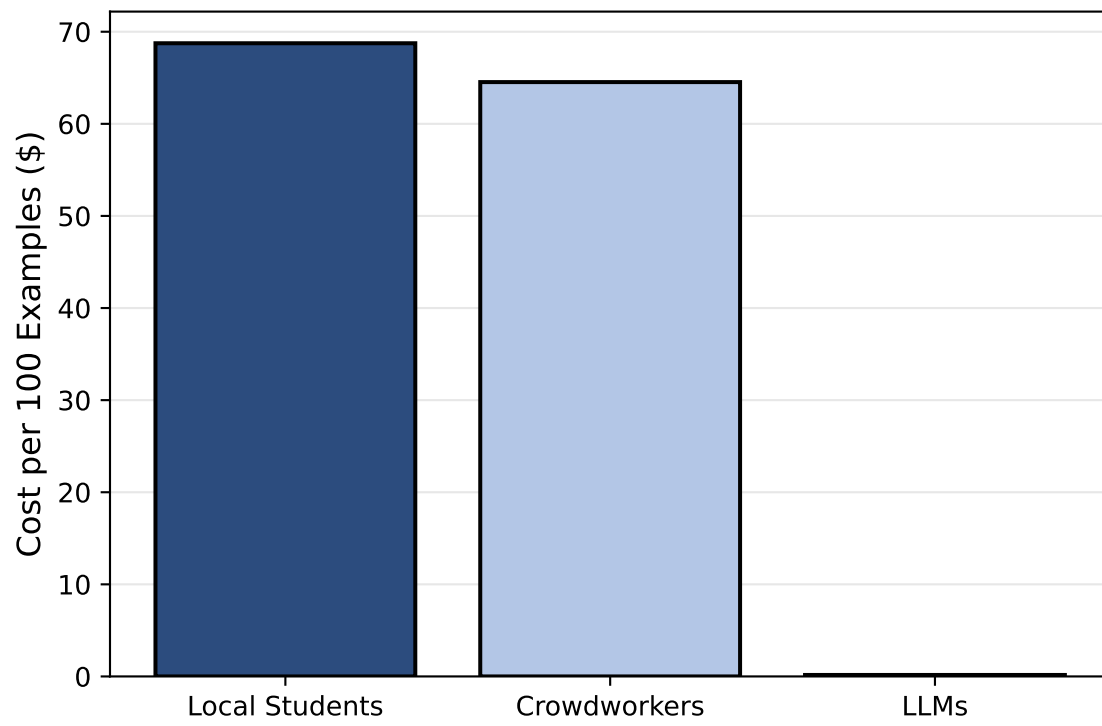
Since I care about recall of sg. and gen. *they* for the application in later research, the recall for those two should be featured apart for humans & LLMs

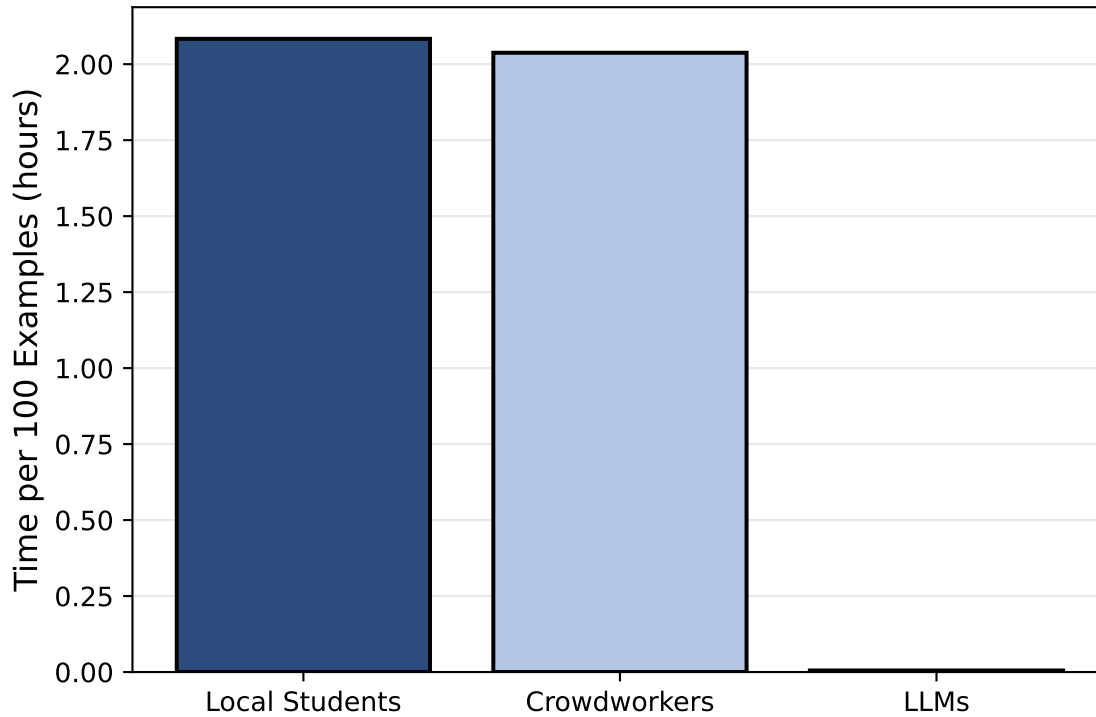
5.3.4 Projection onto naturalistic data

It might make sense to apply the per-class performance measures to a hypothetical test set with naturalistic frequency splits (take from own data) to then generate MCC and get an estimation of how well the different participants would do on naturalistic data. This will address the lingering question if the artificial selection of data points skewed the results or hid anything about the truth

This will be an important point for the conclusion.

5.3.5 Cost





- The cost of crowdworkers was inflated in this study due to filtering inattentive participants only post-hoc and not being able to tell the meaning of “educated in English language” before carrying out the study. Appropriate numbers were obtained thanks to the culling. This means in turn that the responses of 56.1% of crowdworkers were disregarded and their cost was spent fruitlessly.

6 Discussion

6.1 L1 Crowdworkers vs Local L2 Students

The results showed that this task in particular is not automatically solved better by native speakers than by informed L2 speakers. The latter performed at 81.0% as opposed to the 68.0% achieved by native speakers. This is not solely due to the L1 vs L2 status but also a question of competence. The local students had been selected based on their previous high academic achievements and specific knowledge of English linguistics of gender. The difference in performance is not, however, statistically significant. Still, it shows that at this level the native speaker status should no longer be taken as a prerequisite for (annotation) work in this and similar fields or activities. Local students are also preferable due to ethical and practical reasons as communication is easier and there are better mechanisms in place to ensure fairness and research ethics. (Alternative AI suggestion for more adequate phrasing: In this sample,

local L2 student pilots showed similar or higher mean accuracy than non-pilot crowdworkers. Because this comparison is observational and groups differ in selection and expertise, it should not be interpreted as an isolated L1 versus L2 effect. The result instead supports the practical point that trained local student annotators can be competitive with crowdworker annotators in this task. And: These data do not support treating native-speaker status alone as a sufficient criterion for annotator selection in this task.) ## Hypothesis 1

Table 1 reveals that the group of top performers is populated by more humans than LLMs, although most top performances cluster around the 90% mark. To directly address Hypothesis 1, the top performer is not an LLM but indeed a human. This participant was furthermore from the pilot participants recruited from the local student body. Hypothesis 1 is therefore rejected.

6.2 Hypothesis 2

The second Hypothesis aims to test if LLM accuracy at the task outperforms the mean human accuracy. As the overall goal of this research is to assess if LLMs could plausibly replace proper student assistants, it makes sense to define “mean human accuracy” more closely. The participants were recruited more liberally than a standard selection process for a student assistant position. They were recruited from only two seminar groups and were prioritized by availability. The crowd workers self-selected by ticking a box that said they had an education in “English language”, without specifying their level of expertise and, in some cases, not actually having studied linguistics, judging by their answers to the screening. To approximate what level of performance one might expect from an actual student assistant, I additionally report a culling-based benchmark variant as sensitivity analysis. This benchmark is defined by removing inattentive annotators and retaining the top half of the remainder. As this benchmark is interpretive and operational and not design-neutral, its results should be interpreted cautiously. still, it may serve as an estimate for what might be “true human performance.” Inattentiveness is formalized in terms of context-dependent performance, i.e. if their performance is at or below 33% when context is necessary to understand an example.

Figure 1 reveals that there is a marked difference between the overall and the culled human mean. This shows once again that crowdworkers as a whole are not a reliable resource for annotation work. The culled human mean, which is supposed to approach the performance level of a specialized student assistant, still falls short of the best three LLMs’ performance. There, it seems that the prompting strategy had only a minor impact, improving performance by just a few percent. The strongest combination of LLM and prompting strategy, {‘label’: ‘Gemini 3 Pro Preview with URL Access’, ‘value’: 0.9233716475095785}, shows a performance that is , however, not significantly better than the culled human mean at $p = 0.480682373046875$, $\alpha = 0.05$ according to a McNemar’s test. Hypothesis 2 can therefore not be confirmed. It is still remarkable that the best result of the study is achieved by taking the majority vote of the culled human group. Applying this to a real-world setting, it suggests that teams of experts are still the strongest at this kind of annotation task.

6.3 Hypothesis 3

Hypothesis 3 aims to show non-inferiority of LLMs at

6.4 Research question 2

In an attempt to assess the problem-solving behaviors qualitatively, three aspects of human and LLM responses were analyzed: the optional comment fields in the human survey, the LLMs’ reasoning steps, and the accuracy of LLMs in determining context-dependency in the items.

Human-supplied comments were analyzed to reveal the strengths and weaknesses of their reasoning. The following themes were identified: the biggest issue was their understanding of the in-between category of generic *they*, followed by some minor cognitive barriers. Their biggest strength seemed to be their skill in discerning when they needed additional context to make a decision (though the quantitative results show that this did not give them an edge over LLMs). The cognitive barrier for humans in this task lies in the processing of very long input texts. While most items in the dataset are fewer than ten sentences long, item 23 is 38 sentences long and the pronoun in question is located in the penultimate sentence. On top of that, the text uses several uncommon words and therefore requires additional research for most. Human participants, potentially also fatigued when they encounter this item, have to choose how much context they read. In all likelihood, they do not want to read such a lengthy text. LLMs, on the other hand, are much less sensitive to the length of text, as long as the input is well within their window (#SOURCE). Indeed, humans answered wrong N times while LLMs only M (#TODO). As previously discussed (#REF), the category of generic is neither plural nor singular. It is not taught as standard grammar teaching to laypeople and not even linguists are assured to have come across it in their studies or work. Thus, some items in the dataset caused trouble for the human participants. For instance, item 20 shows *they* referring to hypothetical football players, a plural referent. However, the hypothetical nature of the statement seemed more salient to the participants than the plural number of the referent, leading them to classify it as generic. Similarly, item 134 shows *they* referring to *your ancestor*, which is an impersonal use of the second person possessive pronoun *yo*, thus making the reference generic. Most human reasonings focus on the use of *your* but fail to understand its impersonal meaning. The biggest issue with the generic category for human participants were the cases of *they* referring to *everyone* and *noone*. According to the annotation instructions they were to annotate these as plural. The instructions explicitly referred to *everyone*, *noone*, and *nobody* for the plural category. Instead, human participants seemed to have retained the instruction that statements “that [apply] to all X” and accordingly classified these items as generic. This reveals a crucial difference between human and LLM annotators. The latter famously excel at repeating patterns and are thus strong at recognizing cases they received instructions for, regardless of any potential weaknesses in the internal logic of the instructions. Humans, on the other hand, seem to grasp the underlying logic of instructions and override

inconsistent instructions. Had the participants been in a real-world annotation setting, they would have likely asked for clarification on the instructions, thus revealing this inconsistency and helping to improve my annotation scheme. Their quantitative weakness (i.e. incorrectly annotating examples involving *everyone* and *noone* according to the instructions) is thus really a strength, i.e. because they understood the faults of the underlying logic. Though correct usage of an LLM may have also brought this weakness to the forefront.

When analyzing the LLMs' reasoning steps, #TODO

- problems unique to LLMs:

** maybe umkao example and how LLMs made more base assumptions in that case

- strengths unique to LLMs:

Some errors were shared among human and LLM participants, #TODO

To determine the LLMs' accuracy in identifying context-dependency, the prompting strategy that includes the context URL was modified to additionally ask the LLM to declare whenever it decided to access the

6.5 Limitations

Because inferential analyses were primarily based on aggregated accuracies rather than a full item-level hierarchical model, residual item-difficulty effects may remain.

7 Conclusion

The results are mixed and it is not straightforward to say if LLMs have truly attained human level at this particular task.

- Results show that semantically complex annotation in linguistics may be done by LLMs at much cheaper cost. The human-machine margin in cost arguably more than compensates the margin in quality.
- The results also show that the under-the-hood reasoning of LLMs has already diminished the importance of prompt engineering for this task, further reducing the risk of human variability and reducing the importance of human expertise altogether.
- Potentially applicable to any other variable dependent on situational, pragmatic, ... context, i.e. 'distant reading'
- Similar research tends to confirm this finding.

- However, some cursory analysis of LLM errors shows that LLMs are prone to being fooled by context.
- Also, bias towards standard English still a limitation, i.e. innovative forms like specific singular *they* at disadvantage.
- Only tested large commercial LLMs, a limitation common in the literature so far.
- To do:
 - – Further qualitative analysis of human & LLM responses
 - – Application and evaluation of open source LLMs (replicability!), N-shot prompting, fine-tuning
 - – Analyze uses of they in various corpora, potentially uncovering the history of its change and diffusion – – Threat to entry-level / student assistant positions?
 - – Environmental impact concerns?

7.1 Debugging etc.

References

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